

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering **ASSESSMENT TOOL 2 – Case Study**

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	<i>Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)</i>		
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Section / Batch:	B. Tech AI & ML	Date:	12-08-2026

Code of Conduct

I, S. Susannal Devapriyam (192525398) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Susannal Devapriyam

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

CASE STUDY

Feature Matching under Illumination Changes

Computer Vision – Feature Detection, Description and Matching

Given Case Study Statement:

Two images of the same scene are captured under different lighting conditions, causing feature matching errors. The task is to analyze why matching fails and explain how suitable preprocessing and robust feature descriptors can improve matching performance.

Executive Summary:

Feature matching identifies corresponding points between images. When illumination changes, the same physical feature can become brighter, darker, shadowed, highlighted, or lower in contrast. These changes alter local image measurements and can cause conventional matching methods to reject correct correspondences or accept incorrect ones.

A practical solution is to combine controlled illumination-aware preprocessing with robust local feature descriptors. SIFT is a strong classical choice because it uses local gradient structure and provides useful robustness to scale, rotation, and moderate illumination changes. Ratio testing and geometric verification can further remove ambiguous and false matches.

Objectives:

- Understand illumination effects on feature detection and description.
- Identify causes of false, missed, and ambiguous matches.
- Use suitable preprocessing to reduce lighting-related appearance differences.
- Select robust descriptors for stable feature representation.
- Design and justify a reliable matching pipeline.

1. Case Analysis and Interpretation

1.1 Problem Interpretation

The physical scene is assumed to remain substantially the same, while its visual appearance changes because of lighting. Feature matching depends on visual measurements, so a change in illumination can change the numerical representation of the same scene point. The system must distinguish genuine scene differences from illumination-induced differences.

1.2 Feature Matching Process

1. Detect distinctive locations such as corners, blobs, or textured regions.
2. Describe each local neighbourhood using numerical features.
3. Compare descriptors from both images to find candidate correspondences.
4. Reject weak or ambiguous matches using suitable matching tests.
5. Verify the remaining matches using geometric consistency.

1.3 Main Illumination Changes

- Global brightness changes across the entire image.
- Local shadows hide or weaken texture.
- Contrast changes alter local gradients.
- Highlights and reflections create saturated or newly bright regions.
- Exposure differences change the recorded intensity distribution.
- Different light sources can alter colour appearance.
- Non-uniform lighting affects different parts of the scene differently.

1.4 Why Matching Fails

Raw intensity values are not stable under illumination changes. A feature that is distinctive in a bright image may have a very different intensity pattern in a dark image. Direct pixel or weak descriptor comparison can therefore produce large distances between true corresponding regions.

1.5 Case Requirement

The desired system should preserve stable structural information such as local edges, corners, texture arrangement, and geometric relationships while reducing the influence of unstable brightness values.

2. Application of Computer Vision Concepts

2.1 Feature Detection

Feature detection identifies repeatable and distinctive image locations. Corners, blobs, and textured regions are useful because their local structure provides more information than a single pixel. Illumination can nevertheless change feature strength, so robust detection is important.

2.2 Feature Description

A descriptor summarizes the neighbourhood around each keypoint. Gradient-based descriptors are often more robust than raw pixel patches because they emphasize local structure and orientation rather than absolute brightness.

2.3 Robust Descriptor Choices

- SIFT: gradient-based local descriptors with strong robustness to scale, rotation, and moderate illumination changes.
- ORB: fast binary descriptors suitable for real-time systems where computational efficiency is important.
- SURF: another local feature approach designed for efficient matching, subject to implementation availability.
- Normalized local descriptors: reduce the influence of overall intensity and contrast changes.

2.4 Matching Distances

Floating-point descriptors such as SIFT can be compared using Euclidean distance. Binary descriptors such as ORB are commonly compared using Hamming distance. A nearest-neighbour result alone is not sufficient because unrelated regions can also have similar descriptors.

2.5 Ratio Test

The nearest-neighbour ratio test compares the distance of the best candidate with the second-best candidate. A clearly better first candidate is more distinctive, while a small difference between the first and second candidates indicates ambiguity.

2.6 Geometric Verification

After descriptor matching, a robust geometric method such as RANSAC can identify a transformation supported by many correspondences and reject spatially inconsistent matches. This adds an important layer of protection against false matches.

3. In-Depth Analysis of Matching Failure

3.1 Failure Chain

Illumination changes alter intensity and contrast; these changes modify local gradients; modified local information changes descriptors; descriptor distances become less reliable; the matcher then misses correct matches or accepts incorrect ones.

Error Type	Cause	Effect
False match	Unrelated regions appear similar	Incorrect correspondence is accepted
Missed match	True feature appearance changes strongly	Correct correspondence is rejected
Ambiguous match	Repeated or low-texture regions	Several candidates appear equally likely
Unstable keypoint	Feature becomes weak under new lighting	Feature is not detected consistently
Geometric outlier	Incorrect descriptor match survives filtering	Estimated transformation becomes unreliable

3.2 Shadows and Highlights

Shadows can remove visible texture and weaken local gradients. Highlights can saturate pixels and change the appearance of feature regions. Because these effects can be spatially uneven, a single global brightness correction may not solve every mismatch.

3.3 Low Contrast

When an object and its background have similar intensity, feature detection becomes difficult. Contrast normalization may improve visibility, but excessive enhancement can amplify noise and create artificial structures.

3.4 Importance of Geometric Consistency

Correct matches between overlapping views should follow a consistent spatial relationship. RANSAC can estimate a model from candidate correspondences and classify points that do not support the model as outliers.

3.5 Overall Analysis

The case is therefore not simply a nearest-neighbour matching problem. Reliable matching requires suitable preprocessing, repeatable feature detection, robust descriptors, ambiguity rejection, and geometric verification.

4. Solution Design and Justification

4.1 Proposed Solution

Use a multi-stage illumination-aware feature matching pipeline. The goal is to reduce unnecessary lighting variation while preserving stable local structure, then use robust descriptors and verification to obtain reliable correspondences.

Stage	Purpose
Input Images	Read overlapping views of the same scene
Preprocessing	Reduce avoidable illumination and contrast differences
Feature Detection	Find repeatable local structures
Robust Description	Represent stable local gradient or binary structure
Descriptor Matching	Find candidate correspondences
Ratio + RANSAC	Remove ambiguous and geometrically inconsistent matches
Verified Matches	Retain reliable scene correspondences

4.2 Preprocessing Strategy

- Convert to grayscale when colour is not essential to reduce colour-illumination dependence.
- Use contrast normalization when brightness differs strongly.
- Use histogram equalization for suitable global contrast problems.

- Use CLAHE when illumination varies locally and controlled local enhancement is needed.
- Apply mild denoising if enhancement would otherwise amplify noise.

4.3 Descriptor Strategy

SIFT is recommended when robustness is the main priority. ORB is a practical alternative when real-time speed and lower computational cost are important. The descriptor should be selected according to the application rather than assuming one method is best for every case.

4.4 Matching and Verification

- Use Euclidean distance for SIFT-type floating-point descriptors.
- Use Hamming distance for ORB binary descriptors.
- Apply the ratio test to reject ambiguous candidates.
- Use RANSAC-based geometric verification to reject spatially inconsistent correspondences.

4.5 Justification

The proposed solution is feasible because each stage is established in Computer Vision and can be implemented using common image-processing libraries. The pipeline is modular, so preprocessing strength, descriptor choice, and matching thresholds can be adjusted for different lighting conditions.

5. Implementation Plan and Expected Results

5.1 Step-by-Step Implementation

1. Read both images and confirm that the scene overlap is sufficient.
2. Inspect brightness, contrast, shadows, highlights, and noise.
3. Convert to grayscale when colour is not required.
4. Apply controlled normalization or CLAHE if illumination differences are significant.
5. Detect keypoints using SIFT or another suitable robust detector.
6. Compute descriptors around the detected keypoints.
7. Match descriptors using the appropriate distance measure.
8. Apply the nearest-neighbour ratio test.
9. Estimate geometric consistency using RANSAC and reject outliers.
10. Visualize and evaluate the final verified correspondences.

5.2 Expected Behaviour

Condition	Before Improvement	After Proposed Pipeline
Brightness difference	Descriptor distances become unstable	Preprocessing reduces avoidable variation
Shadows	Some features become weak or disappear	Robust features improve repeatability
Highlights	Unreliable saturated regions create errors	Filtering and verification reduce false matches
Repeated texture	Several candidates appear similar	Ratio testing reduces ambiguity
False matches	Incorrect pairs may survive	RANSAC rejects inconsistent pairs
Final matches	Unstable or discontinuous	More reliable and spatially consistent

5.3 Evaluation Measures

- Number of detected keypoints in each image.
- Number of initial descriptor matches.
- Matches remaining after ratio testing.
- Number and percentage of RANSAC inliers.
- Spatial distribution and visual quality of final matches.
- Stability across several illumination conditions.

5.4 Limitations

- Aggressive enhancement may amplify noise.
- No descriptor is perfect under every lighting and viewpoint condition.
- SIFT can require more computation than lightweight binary descriptors.
- Very strong shadows or saturation can remove information that preprocessing cannot recover.
- Repeated patterns can remain ambiguous even with robust descriptors.

5.5 Expected Outcome

The expected result is a higher proportion of correct correspondences, fewer false matches, and a more spatially consistent set of matches that can support image registration, object localization, panorama creation, and related Computer Vision tasks.

6. Conclusion

6.1 Conclusion

Feature matching under illumination changes is difficult because the same physical feature can produce different intensities, gradients, contrast patterns, and local appearances. These changes can cause feature detectors to miss keypoints and can make corresponding descriptors appear less similar. Simple intensity-based matching can therefore fail.

A robust response is to combine controlled illumination-aware preprocessing with robust local descriptors. Ratio testing reduces ambiguous matches, while geometric verification rejects correspondences that do not agree with the scene geometry. Together, these stages improve matching stability without depending only on raw pixel similarity.

6.2 Key Recommendations

- Use preprocessing only as strongly as necessary.
- Prefer structure- or gradient-based descriptors when illumination varies.
- Use SIFT when robustness is prioritized and ORB when computational efficiency is important.
- Apply ratio testing to reduce ambiguous correspondences.
- Use RANSAC or another geometric verification method before accepting final matches.
- Evaluate the pipeline using numerical and visual evidence.

6.3 Final Statement

Therefore, reliable feature matching under changing illumination should combine illumination-aware preprocessing, robust local feature descriptors, reliable descriptor matching, and geometric verification. This integrated design directly addresses the causes of matching failure and improves the usefulness of correspondences for real-world Computer Vision applications.

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